

2021 International Energy Conservation Code (IECC) - Commercial Provisions

Section C402 - Building Envelope Thermal Performance

Commercial building thermal envelopes in high cooling degree-day climate zones (Climate Zones 1, 2, and 3) must meet stringent thermal resistance standards.

Continuous insulation (ci) is mandatory across above-grade opaque mass walls and steel-framed wall assemblies to prevent conductive heat flux during peak solar radiation.

Section C402.1.4 - Fenestration and Solar Heat Gain Coefficient

Vertical fenestration and glazing assemblies must maintain maximum Solar Heat Gain Coefficients ($SHGC \leq 0.25$) and low U-factors to minimize radiant heat ingress.

Mitigation of thermal bridging across building envelope joints is required to eliminate localized structural hotspots.

Section C403 - Mechanical System Modulation & Economizers

HVAC equipment must include automated air and water economizers to leverage night-time ambient cooling.

Demand-responsive temperature controls must be capable of automatically resetting cooling setpoints during peak urban thermal emergencies.